QP CODE 25900509

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Core Course (DSC) - MG3DSCMAT203 - ESSENTIALSOF QUANTITATIVE DECISION MAKING
(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Part A

(Answer any 5 questions , Each question carries 2 marks)

1. H.M (Formula) ……..1mark
   Average speed (Harmonic mean) = 40 km/h ……1mark [A] / [CO1

2. Standard deviation (formula) ……..1mark
   S.D = 4.08 ……..1mark [A] / [CO1
3. There is no correlation (or very weak correlation) between the two variables .
   . ……2marks [U] / [CO2]
4. 2X- 4Y-5 = 0 is the regression equation of y on x
   Y = $\frac{2}{4}$.X - $\frac{5}{4}$ … ……1mark

   Regression coefficient , byx = ½ ……..1mark [U] / [CO2]

5. Solution of Lpp ……..1mark

   Feasible solution of LPP ……..1mark [U] / [CO3]

6. Any two examples of surplus variable ……..1 mark each [A] / [CO3]

7. The main use of the MODI method in transportation problems is to test the optimality of an initial basic feasible solution and improve it (if necessary) to find the minimum transportation cost ……2marks [U] / [CO4]

8. Explain the North - West Corner method ……2marks [K] / [CO4]

Part B

(Answer any 5 questions, Each question carries **6** marks)

9. Median = 40 ……..2marks
   M.D = $\frac{1}{N}\sum|x - Md|f$ ……1mark
   $\sum$ |x−M_d|. f = 620 ……..2marks
   M.D = $\frac{620}{60}$ =10.33 ……1mark [A] / [CO1]

10. Median (formula)

$$\text{Median} = l + \frac{\frac{N}{2} - m}{f} \times c$$

……..1marks

Substituting values in the formula ……2marks

Calculations of missing frequency , **f = 25** ……3marks [A] / [CO1]

11. Explain any 3 types of correlation ……..2marks each [U] / [CO2]
(Positive Correlation, Negative Correlation, Zero (or No) Correlation , Linear Correlation, Non-linear Correlation etc.)

12. Karl Pearson's Coefficient of Correlation

$$r = \frac{n.\sum xy - \sum x.\sum y}{\sqrt{n.\sum x^2 - (\sum x)^2} \cdot \sqrt{n.\sum y^2 - (\sum y)^2}}$$

.Any correct formula ……..1 mark

Calculations …….. 3marks

$\sum X = 12$, $\sum Y = 32$, $\sum X^2 = 34$, $\sum Y^2 = 228$, $\sum XY = 87$

Coefficient of Correlation .**r = 0.93**……..1mark

Comment : Since r=0.93, it indicates a strong positive correlation between Series X and Y (or if the values of X increase, then values of Y also tend to increase.)
. ……..1mark [A] / [CO2

13. Write the general form of a Linear Programming Problem

Objective function …… 2marks

Constraints ……3 marks

Non negativity condition …….1 mark [K] / [CO3]

14. Graph and feasible region ……. 2marks

Values of the objective functions at the vertices of the feasible region:

Z(0,0)=0, Z (0,40) =40, Z(30,0)=240, Z (20,20)=180 ……..2 marks

Solution: X= 30 ,Y=0 **; Max Z = 240** ……2marks [A] / [CO3]

15. Initial solution by Least Cost Matrix method

Solution : X_{12}= 12, X_{13}=1 , X_{14}=9, X_{23}=15, X_{31}=7, X_{33}=1

Total transportation Cost = 150. …….6marks [A] / [CO4]

16. Objective Function:
Min. Z = $6.X_{11}+1.X_{12}+2.X_{13}+5.X_{21}+2.X_{22}+3.X_{23}+1.X_{31}+4.X_{32}+5.X_{33}$……2marks

Sub to. the constraints

$X_{11}+X_{12}+X_{13}=6$
$X_{21}+X_{22}+X_{23}=4$
$X_{31}+X_{32}+X_{33}=10$ and $X_{11}+X_{21}+X_{31}=2$
$X_{12}+X_{22}+X_{32}=3$
$X_{13}+X_{23}+X_{33}=15$ ……3marks

Non Negativity Condition Xij ≥ 0, i,j =1,2,3 ……1marks [U] / [CO4]

Part C

(Answer any 3 questions, Each question carries 10 marks)

17. a) Total weekly wages:

Factory A= 345×450=155,250

Factory B= 290×530=153,700

Factory A pays the larger total weekly wages (₹155,250) …….3marks

b) C.V=$\frac{S.D}{Mean}$*100

C.V$_A$ = $\frac{25}{345}$ *100 = 7.25%; C.V$_B$ = $\frac{45}{290}$*100= 15.52%

Factory B shows greater variability in the wage rate …….3marks

c) Combined Mean $\bar{X} = \frac{n1.\overline{XA}+n2\overline{XB}.}{n1+n2}$ = 315.26 ……. 2marks

Combined Standard Deviation

S.D = $\sqrt{\frac{n1.s1^2+n2s2^2+n1d1^2+n2d2^2}{n1+n2}}$ = 46.18 …….2marks

18. a) Regression equation of Y on X : $(Y-\bar{Y}) = b_{yx}(X-\bar{X})$ ……1mark

$\bar{X} = \frac{\Sigma X}{n}$, $\bar{Y} = \frac{\Sigma Y}{n}$ $b_{yx} = \frac{n.\sum xy - \sum x.\sum y}{n.\sum x^2-(\sum x)^2}$ ……..2marks

Calculations

$\bar{X} = \frac{\Sigma X}{n}$ = 50 $\bar{Y} = \frac{\Sigma Y}{n}$ = 100.3 , byx = -0.1383 ……2marks

Regression equation of **Y on X**: Y= −0.1383.X + 107.21 ……1marks

b) Regression coefficient of X on Y:

$b_{xy} = \frac{n.\sum xy - \sum x.\sum y}{n.\sum y^2-(\sum y)^2}$ = - 0.1114

Correlation coefficient: r = $\sqrt{byx.bxy}$ = - 0.124 ……. 3marks

(Give mark for direct calculations also)

c) Estimated blood pressure for a 45-year-old:= −0.1383*45 + 107.21

=100.99 …… 1mark. [A] / [CO2]

19. Spearman's rank correlation coefficient (for tied ranks) $\boldsymbol{\rho = 1 - [\frac{6.\sum D^2 + \frac{\sum m(m^2-1)}{2}}{n.(n^2-1)}]}$

……..2marks

Assigning ranks ……3 marks

Calculations : $\sum D^2 = 45$ and $\frac{\sum m(m^2-1)}{2}$ **= 3+3+12=18 ……..3 marks**

Spearman's rank correlation coefficient (for tied ranks) $\boldsymbol{\rho = 1 - \frac{6*45+18}{8*63}}$ **= 0.429 ….2marks**

[Give 9 marks for using the formula, $\rho = 1 - \left[\frac{\sum D^2}{n.(n^2-1)} 6\right]$ = **0.46]** [A] / [CO2]

20. a) Decision variables : x1= number of chairs produced per day and x2= number of tables produced per day.

Objective function: Max Z= 50x1+100x2

Sub to.

$5x1+7x2 \leq 20$

$8x1 \leq 30$

$x1, x2 \geq 0$ …….5marks

b) Graph and feasible Region ……..3marks

Solution : **Max Z= 2000/7= 285.71**

x1 = 0 and x2 =.20/7=2.86 …….....2marks [An] / [CO3]

21. Standard form……..2marks

Initial B.F.S …….2 marks

First simplex table……..2marks

Second simplex table……2marks

Solution : Max Z= 10 ; x1=2 and x2= 0 …….2marks [A] / [CO3]

22. a) Initial solution by NWC……5marks

Solution: X_{11}=15, X_{21}=3 , X_{22}=22, X_{32}=3, X_{33}=22

X_{34}= 5 (Dummy) and X_{44}= 20 (Dummy)

Total transportation cost = **442**

b) Initial solution by VAM……5marks

Solution : X_{12}=15 , X_{21}=18 , X_{23}=2 , X_{32}=10, X_{33}=20 ,

X_{24} =5 (Dummy) and X_{44}=20 (Dummy)

Total transportation cost =**241** [A] / [CO4]NB:

**Stage wise marks for any correct alternate method can be fixed by the examiner